Popularising Summary

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Master's thesis: an extremal problem on connectivity in networks (Erdős Problem 915).

Many of the networks we rely on are useful precisely because they offer backup routes. If one road is closed, traffic can take another. If one cable fails, a message can still reach its destination along a different chain of links. Mathematicians study this kind of structure with graphs, where dots stand for objects and lines stand for the connections between them. This thesis asks a simple-sounding question about such networks: how many connections can a network with a fixed number of points have, if we forbid any two points from being joined by too many separate, non-overlapping routes?

That question is a version of a problem posed decades ago by the mathematicians Béla Bollobás and Paul Erdős, listed today as Erdős Problem 915. In its original form it concerns ordinary networks and routes that are not allowed to share a connection. For that case the answer has been known since a theorem of Mader. This thesis starts from that result and then changes the rules of the network one at a time: connections may be doubled up, directions may be added so that a link works only one way, or a single connection may join more than two points at once. With each change the pencil-and-paper proof becomes longer and more delicate, until checking it by hand stops being realistic.

At that point the thesis hands the work to a computer. It builds a single program that can describe every version of the network in the same way, measure exactly how many separate routes run between any two points, and prove the answer outright for small networks by checking every possibility. For the larger networks, where checking everything is hopeless, the program instead searches for good examples by imitating the way hot metal cools and slowly settles into an orderly shape, drifting at first and then locking into place. Reassuringly, when this search is pointed at the cases we already understand, it rediscovers the very networks the earlier proofs had singled out. The interplay between proof and experiment then pays off. For networks whose connections may each tie several points together at once, the thesis determines the exact largest possible network under the two strictest limits on backup routes, whatever the total number of points and however many points each connection joins. For the most stubborn version, the one with one-way links, it shows that the whole infinite family of questions reduces to a single finite check on networks of one fixed size, so that one exhaustive computer check at that single size would settle every larger case at the same time. That final check is the main question the thesis leaves open.